

CURRICULUM VITAE

Summary Statement

I am a Data Scientist at Mayo Clinic developing and deploying generative AI systems to advance clinical care and biomedical research. I completed my PhD in Biomedical Informatics at the University of Pittsburgh School of Medicine under the mentorship of Dr. Rafael Ceschin. My thesis focused on applying network science, machine learning, and neuroimaging methods to predict long-term neurodevelopmental outcomes in children with congenital heart disease. My background spans software engineering in the telecommunications industry, translational research in academic medical centers and nonprofit settings, and applied data science in clinical environments. My work integrates data science and software engineering to enable clinically meaningful predictive modeling, computer-aided detection, and intelligent decision-support tools in medicine.

BIOGRAPHICAL

Name:	Joy Roy	Citizenship:	USA
Affiliation:	Mayo Clinic	Website:	joyroy.org
Location:	Remote (Baltimore, MD)	E-Mail Address:	roy.joy@mayo.edu

EDUCATION and TRAINING

GRADUATE:

Dates Attended	Name and Location	Degree Received	Subject
2020 - 2024	University of Pittsburgh School of Medicine Pittsburgh, PA	PhD	Biomedical Informatics

UNDERGRADUATE:

Dates Attended	Name and Location	Degree Received	Subject
2015 - 2019	University of Maryland, Baltimore County Baltimore, MD	BS	Major: Bioinformatics and Computational Biology
2015 - 2019	University of Maryland, Baltimore County Baltimore, MD	BA	Major: Mathematics Minor: Computer Science

APPOINTMENTS and POSITIONS

Senior Data Scientist

Mayo Clinic

2025 - Present

Remote - Baltimore, MD

- Design, develop, and evaluate generative and agentic AI systems to support clinical decision-making and improve healthcare delivery.
- Lead clinical research and validation studies, including IRB protocol development, study design, statistical analysis, and assessment of model performance, safety, and bias/fairness.
- Collaborate with physicians, multidisciplinary research teams, and industry partners (e.g., Microsoft) to implement and evaluate AI-driven clinical tools in real-world settings.

Health Informatics Scientist

Henry M Jackson Foundation

2025 - 2025

Bethesda, MD

- Appointed with the National Intrepid Center of Excellence at Walter Reed Hospital (WRNMMC)
- Designed a data ingestion, harmonization, and storage framework to centralize blast traumatic brain injury data for machine learning applications.
- Support blast traumatic brain injury research using computer science approaches.

Pre-Doctoral Fellow

Pediatric Imaging Research Center, Children's Hospital of Pittsburgh

2020 - 2024

Pittsburgh, PA

- Collaborated with Department of Psychiatry, Radiology, and Biomedical Informatics to do high level data analyses on clinical data
- Developed, trained, and evaluated machine learning models, including Random Forests and Convolutional Neural Networks, to derive predictive insights from large medical imaging datasets.
- Automated MRI preprocessing pipelines with custom Python scripts, utilizing libraries such as FSL and NumPy to perform tasks like segmentation, registration, and data normalization.
- Developed new open source commandline software to accelerate research

Software Engineer

Hughes Network Systems

2019 - 2020

Gaithersburg, MD

- Designed and implemented real time software for protocols, algorithms, and products using C/C++
- Maintained Jenkins pipelines for automated unit testing and continuous integration and continuous development.
- Handled ITAR controlled technical data

Bioinformatics Research Assistant

Lobo Lab, University of Maryland Baltimore County

2017 - 2020

Baltimore, MD

- Developed an algorithm to encode and curate the biological data in planaria based on anatomical ontologies
- Lead a project team to develop a [software](#) tool written in C/C++ (with Qt framework) integrating this methodology with a GUI for the curation of data into a database managed in SQL

ORISE Research Fellow

Food and Drug Administration

Summer 2017

Silver Spring, MD

- Conducted research on Fc receptors and their effect on therapeutic monoclonal antibodies.
- Digested and isolated antibody fragments, used Flow Cytometry, performed Western Blot analysis, gel electrophoresis, and Mixed Lymphocyte Reactions.

HONORS

2024	Finalist, Biomedical Image of the Year MRI Preprocessing Complete University of Pittsburgh School of Medicine Biomedical Graduate Student Association
2023	Runner-Up, Best Student Paper of the Year, Department of Biomedical Informatics Limbic pathway vulnerability associates with neurologic outcome in children after cardiac arrest <i>Resuscitation</i> , 2023 Jan;182:109634
2023	Bronze Medal Focus Talk Functional Network Organization is Atypical in Congenital Heart Disease National Library of Medicine T15 Training Conference 2023 - Stanford University
2022	Runner-Up, Student Paper of the Year, Department of Biomedical Informatics Defining telehealth for research, implementation, and equity <i>Journal of Medical Internet Research</i> , 2022 Apr 13;24(4):e35037
2019	UMBC Faculty Award for Excellence in Bioinformatics University of Maryland Baltimore County
2019	Departmental Honors in Research, Department of Biology University of Maryland Baltimore County

FELLOWSHIPS AND AWARDS

2020-2024	National Library of Medicine T15 Training Grant [5T15-LM007059-35] Role: Pre-doctoral Fellow Department of Biomedical Informatics University of Pittsburgh School of Medicine
2017-2018	Undergraduate Research Award Role: Student Investigator Office of Vice President for Research University of Maryland Baltimore County

PROFESSIONAL ACTIVITIES

Seminars and Lectures

2022 UPMC Hillman Cancer Center Academy
Pittsburgh, PA
Intro to Imaging Informatics Research

Graduate School Teaching

2022 Teaching Assistant: Publication and Presentations
Department of Biomedical Informatics,
University of Pittsburgh School of Medicine, Pittsburgh, PA

Student Mentoring

2023 Jose Maldonado, CoSSBI Summer Scholar
2022 Ansh Goyal, CoSSBI Summer Scholar

Volunteering

2022-2024 Pitt DBMI Student Representative
2023 Pitt BGSA Symposium Planning Committee Member
2021-2024 Pitt BGSA Program Representative
2021-2023 Pitt BGSA Big Buddy

PRESENTATIONS

1. Roy J, Reynolds W, Wallace J, Badaly D, Ceschin R. Exploring the Impact of Network Construction Methods and Threshold Selection on Graph Analysis Outcomes. Poster presented at: Flux Society Annual Meeting; 2024; Baltimore, MD.
 2. Roy J, Reynolds W, Wallace J, Badaly D, Panigrahy A, Ceschin R. Functional network organization is locally atypical in congenital heart defect. Lecture presented at: Pittsburgh MR Physicists Monthly Seminar; 2024; Pittsburgh, PA.
 3. Roy J, Reynolds W, Frank M, Panigrahy A, Ceschin R. Functional network organization is globally atypical in congenital heart defect. Poster presented at: Flux Society Symposium; 2023; Santa Rosa, CA.
 4. Roy J, Reynolds W, Frank M, Panigrahy A, Ceschin R. Functional network organization is globally atypical in congenital heart defect. Oral presented at: NLM T15 Training Conference; 2023; Stanford University.
 5. Roy J, Ceschin R, Devine D, Panigrahy A, Fink EL. Diffusion tensor imaging as a biomarker for pediatric cardiac arrest outcomes. Poster presented at: American Medical Informatics Association Symposium; 2022; Washington, D.C.
 6. Roy J, Ceschin R, Devine D, Panigrahy A, Fink EL. Diffusion tensor imaging as a biomarker for pediatric cardiac arrest outcomes. Oral presented at: NLM T15 Training Conference; 2022; University of Buffalo School of Medicine.
 7. Roy J, Panigrahy A, Ceschin R. Evaluating brain age models in adolescents with CHD. Poster presented at: American Medical Informatics Association Symposium; 2021; San Diego, CA.
 8. Roy J, Cheung E, Bhatti J, Muneem A, Lobo D. Semi-automatic ontology curation methods for a planarian gene expression pattern database. Oral presented at: Undergraduate Research and Creative Achievement Day; 2019; University of Maryland Baltimore County.
 9. Roy J, Cheung E, Bhatti J, Muneem A, Lobo D. A software tool for the curation of planarian gene expression patterns. Oral presented at: National Symposium for Undergraduate Research; 2018; St. Jude Hospital.
 10. Roy J, Hess E, Chowdhury M, Swisher J, Feldman G. The functional capabilities of an IgG4 FC are comparable to its IgG1 analog. Poster presented at: Annual Biomedical Research Conference for Minority Students; 2017; Phoenix, AZ.
 11. Roy J, Hess E, Chowdhury M, Swisher J, Feldman G. The functional capabilities of an IgG4 FC are comparable to its IgG1 analog. Poster presented at: Food and Drug Administration Summer Research Day; 2017; Silver Springs, MD.
-

PUBLICATIONS

1. Roy J, Reynolds W, Wallace J, Badaly D, and Ceschin R. Functional network organization is locally atypical in children, adolescents, and young adults with congenital heart disease. *NeuroImage Clinical*. 2026; 49:103965. DOI: [10.1016/j.nicl.2026.103965](https://doi.org/10.1016/j.nicl.2026.103965)
2. Roy J. Network Analysis of Multimodal MRI to Identify Regional Associations with Neurodevelopmental Outcomes in Children with Congenital Heart Disease. Dissertation. University of Pittsburgh School of Medicine, 2025 Feb 24. 140 pp. Available from: <https://d-scholarship.pitt.edu/47300>
3. Roy J, Reynolds W, Panigrahy A, and Ceschin R. Functional network organization is locally atypical in children and adolescents with congenital heart disease. *medRxiv*. 2024 Apr 20 :2024.04.19.24306106. DOI: [10.1101/2024.04.19.24306106](https://doi.org/10.1101/2024.04.19.24306106). Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11065028/>
4. Roy J, Jarvis JM, Schmithorst V, Lee V, Devine D, Meyers B, Munjal N, Clark RSB, Kochanek PM, Panigrahy A, Ceschin R, and Fink EL. Limbic pathway vulnerability associates with neurologic outcome in children after cardiac arrest. *Resuscitation*. 2023 Jan 1; 182:109634. DOI: [10.1016/j.resuscitation.2022.10.026](https://doi.org/10.1016/j.resuscitation.2022.10.026). Available from: <https://www.sciencedirect.com/science/article/pii/S0300957222007067>
5. Roy J, Levy DR, and Senathirajah Y. Defining Telehealth for Research, Implementation, and Equity. *Journal of Medical Internet Research*. 2022 Apr 13; 24:e35037. DOI: [10.2196/35037](https://doi.org/10.2196/35037). Available from: <https://www.jmir.org/2022/4/e35037>
6. Roy J, Cheung E, Bhatti J, Muneem A, and Lobo D. Curation and annotation of planarian gene expression patterns with segmented reference morphologies. *Bioinformatics*. 2020 May 1; 36:2881–7. DOI: [10.1093/bioinformatics/btaa023](https://doi.org/10.1093/bioinformatics/btaa023). Available from: <https://doi.org/10.1093/bioinformatics/btaa023>

– END –